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Analytical Methods

Simultaneous voltammetric analysis of tetracycline antibiotics in foods

Yongnian Ni [a,b,*], Shuzhen Li [b], Serge Kokot [c]

[a] State Key Laboratory of Food Science and Technology, Nanchang University, Nanchang, Jiangxi 330047, China
[b] Department of Chemistry, Nanchang University, Nanchang, Jiangxi 330031, China
[c] School of Physical and Chemical Sciences, Queensland University of Technology, Brisbane, Queensland 4001, Australia

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ABSTRACT

The new simple, cost effective and well performing differential pulse stripping voltammetry method for the simultaneous determination of antibiotics – oxytetracycline, tetracycline and chlorotetracycline was researched and developed. It depends on the reduction of these compounds at a hanging mercury drop electrode. The samples were extracted from (i) spiked animal feed, and (ii) fresh fish muscle dosed with the drugs. The voltammograms from the drug mixture produced complex, overlapping profiles, and chemometrics methods were applied for calibration modelling. The analytical linear ranges were within 0.02–0.18 μg mL^{-1} and the corresponding LODs were 3–5 μg L^{-1} for the three analytes. These values compare well with those from the HPLC and fluorescence methods in the literature. The % relative prediction errors from the verification trials were between 4% and 9% with % Recoveries being 103–107. Also, the % Recoveries of the antibiotics from animal feeds as measured by the new method were comparable with those from the HPLC analysis (85%), i.e. the method is highly competitive, especially as a screening approach.

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1. Introduction

Quantitative analysis of antibiotics such as the tetracyclines (TCs), oxytetracycline (OTC), tetracycline (TC) and chlorotetracycline (CTC) are important because these drugs are widely used as animal feed additives to promote growth (Kumar, Gupta, Chander, & Singh, 2005). These substances remain as residues in animals, fish and birds, and are harmful to humans at quite low levels (Wegener, 2003). The presence of these residues is non-trivial because the growth promoting effect of the antibiotic is associated with its inhibitory effect on intestinal microbes, which generally leads to a change in the balance of the intestinal flora. The antibiotic residues from the feed additives, through their antimicrobial effects, can diminish the activity of the immune system in the intestinal tract. Although it is often assumed that resistance observed involving animal bacteria is entirely the result of the use of therapeutic substances, it has been suggested that the feeding of antibiotics for growth promotion purposes also contributed. Thus, it has been recommended that antibiotics for nutritional use should be restricted to those, which are of economic value to livestock production, have little or no implication as therapeutic agents for humans or animals, and do not impair the efficacy of therapeutic antibiotics through the development of resistant organisms (Anadon & Martinez-Larranaga, 1999).

In this context, from 2001, the European Union (EU), has taken measures to protect consumers' health in the food safety area and reduce antimicrobial resistance (Commission Regulation, 2002). For example, EU legislation on animal nutrition from 2006 bans the presence of antibiotics for growth promotion in animal feed (Commission Regulation, 2005). Both the US and EU have set maximum residue limits (MRL) for TC, OTC and CTC in fish: 2 μg g^{-1} (sum of the tetracyclines, US), and 100 ng g^{-1} (for each of the TCs, EU) (Web). However, regardless of any legislation, covert usage of antimicrobial agents in food producing animals persists in North America, Australia (Hammerum et al., 2007) and many developing countries, where the antimicrobial resistance problems are particularly pressing (Okeke et al., 2005). Human migration and global food markets may facilitate the spread of antimicrobial resistance around the world (Díaz Granados, Cardo, & Mac Gowan, 2008). This indicates that any meaningful response to the problem requires input from many sciences and technologies, which can contribute to the research and development of novel detection and forensic approaches to combat the problem at source. Specifically, at the food production stage, during transit to the user, the quality of the final product is monitored during the marketing process, on-field or -farm, and finally, at the consumer interface. This work is a contribution in this context and the novel method described, in principle, may be universally applied at any of the mentioned stages.

* Corresponding author at: Department of Chemistry, Nanchang University, Nanchang, Jiangxi 330031, China. Tel./fax: +86 791 3969500.
E-mail address: ynni@ncu.edu.cn (Y. Ni).

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Most methods for the determination of TCs use high performance liquid chromatography (HPLC) (Fritz & Zuo, 2007), thin layer chromatography (Crecelius, Clench, Richards, & Parr, 2002), capillary electrophoresis (CE) (Wei, Deng, Liu, & Zhao, 2003), and microbiological assay (Pellinen, Bylund, Virta, Niemi, & Karp, 2002). Other methodologies based on titrimetry (Nabi, Abu-Nameh, & Helaleh, 1998) and differential pulse polarography (Pinilla Gil, Calvo Blazquez, Garcia-Monco Carra, & Sanchez Misiego, 1989) have been described. Spectrofluorometric methods with pretreatment have been used in the past (Poiger & Schalatter, 1976), but they involve tedious prior extraction steps. Notwithstanding these techniques, the important technique of choice for analysis of antibiotics is high performance liquid chromatography coupled to mass spectrometry (HPLC–MS) (Reverte, Borrull, Pocurull, & Marce, 2003) and tandem spectrometry (Cherlet, Schelkens, Croubels, & De Backer, 2003) because of their superior sensitivity and selectivity. However, these techniques are costly to acquire, run and maintain; they require relatively sophisticated laboratory facilities and specialist operators. Thus, more simple, cost effective, rapid screening technologies are required for global use to select the questionable samples, which could then be sent for confirmatory analysis by HPLC–MS.

Other than the separation methods such as chromatography and CE, only a few others have been described for the simultaneous determination of TCs. Thus, for example, spectrofluorometric techniques combined with multivariate calibration have been proposed to quantify TCs individually. Goicoechea and Olivieri (1999) applied partial least-squares (PLS) and hybrid linear analysis (HLA) which did not require the knowledge of pure-component spectra for spectrofluorometric determination of TC in human serum samples in the range of 0.0–4.0 $\mu g\ mL^{-1}$. The limit of determination (LOD) was about 0.20 $\mu g\ mL^{-1}$. (Garcia, Sarabia, and Cruz Ortiz (2004) applied bilinear and trilinear PLS to determine OTC (40.32–282.24 $\mu g\ L^{-1}$), TC (35.35–232.30 $\mu g\ L^{-1}$) and CTC (39.76–278.32 $\mu g\ L^{-1}$) by the fluorescence technique and their corresponding LODs were 17.75, 9.93 and 26.31 $\mu g\ L^{-1}$, respectively.

Recently, Valverde et al. applied a method using photochemically induced fluorescence signals combined with both first- and second-order multivariate calibrations, PLS (Valverde, Garcia, Galera, & Goicoechea, 2006a, 2006b) and parallel factor modelling (PARAFAC), N-way partial least-squares (NPLS) and bilinear least-squares (BLLS) (Valverde et al., 2006a,b) for analysis of a mixture of three TCs in surface water samples. The linear ranges were 80–4000, 100–4000 and 80–1500 $\mu g\ L^{-1}$ for OTC, TC and CTC, respectively. Their corresponding LOD values were 32, 14 and 5 $\mu g\ L^{-1}$, respectively. The RMSEP (root-mean-square error of prediction) obtained were 35, 95 and 85 $\mu g\ L^{-1}$ for OTC, TC and CTC, respectively (Valverde et al., 2006a,b).

Polarographic and voltammetric methods, such as differential pulse polarography (DPP), differential stripping voltammetry (DSV), and adsorptive voltammetry, generally have high sensitivity and are widely used as alternative methods for micro-analysis in many cases. However, their applicability for the determination of multi-component mixtures, especially for some organic compounds with similar molecular structures, are rather limited, because seriously overlapping polarograms or voltammograms are often obtained. Diverse chemometrics methods, such as multiple linear regression (MLR) (Garcia, Martin, Jimenez, Jimenez, & Arias, 1994), Kalman filter (KF) (Brown & Brown, 1981), partial least-squares (PLS) (Henrion, Henrion, Henrion, & Scholz, 1990), principal component regression (PCR) and iterative target transformation factor analysis (ITTFA) (Ni & Jin, 1999), have been shown to overcome this limitation and have resolved successfully the overlapping bands. These chemometrics approaches have also been applied for the determination of analytes in binary or ternary mixtures of similar organic compounds (Ni, Wang, & Kokot, 2006). Apart from the advantage of high sensitivity, electroanalytical methods, generally, give good precision and accuracy, and have relatively low-costs. Consequently, an analytical method was researched and developed for the simultaneous determination of the OTC, TC and CTC with the use of differential pulse stripping voltammetry (DPSV) and the aid of chemometrics.

2. Experimental

2.1. Apparatus

A BAS 100B/W electrochemical analyser (BAS) equipped with a PAR 303A cell stand (EG&G) was used for voltammetric measurements; the three-electrode cell contained a hanging mercury drop electrode (HMDE) as the working electrode, an Ag–AgCl (BAS MF2020) reference electrode and a platinum wire auxiliary electrode. pH measurements were performed on an Orion SA 720 pH meter equipped with an Ag–AgCl glass combination pH electrode. All experiments were carried out at 25 °C.

HPLC measurements were made with an Agilent 1100 Series HPLC–DAD system, a vacuum degasser, quaternary pump, autosampler, injector with a 100 μL loop, an Agilent ZORBAX Eclipse XDB-C18 column (4.6 mm × 250 mm, 5 μm) together with an Agilent Zorbax high pressure reliance cartridge guard-column (C18, 12.5 mm × 4.6 mm, 5 μm) and a variable wavelength UV visible detector.

2.2. Reagents

OTC, TC and CTC antibiotics were obtained from Sigma (USA). Stock solutions (1 $mg\ mL^{-1}$) of OTC, TC and CTC were prepared by dissolving 10 mg in 10 mL of methanol. They were kept in brown glass vials in a refrigerator and were stable for a maximum of 1 month. Working standard solutions (10 $\mu g\ mL^{-1}$) were prepared daily by dilution of stock solutions in methanol to appropriate concentrations. These solutions were prepared immediately before use. A Britton–Robinson buffer solution (pH 3.78) was prepared by adding 22.5 mL of 0.2 mol L^{-1} sodium hydroxide solution into 100 mL of a mixed acid, containing 0.04 mol L^{-1} of each of boric, ortho-phosphoric and acetic acids. A McIlvaine buffer solution (pH 4.0) was prepared by mixing 0.1 mol L^{-1} citric acid (61.45%) with 0.2 mol L^{-1} disodium hydrogenphosphate (38.55%). All chemicals were of analytical reagent grade and distilled water was used throughout the experiments.

2.3. Sample preparation

2.3.1. Experimental procedure: fish and antibiotics

Healthy, live carp, with approximate equal weight of 500 g, were obtained from the local market. The fish were kept in four outdoor experimental tanks with about 50 L fresh water (three fish per tank), and were used in various experiments with the antibiotics. In each tank, the fish were kept in fresh water and at room temperature (average 25 °C). The fish were acclimatised for 5 days prior to being fed on commercial food in the absence of any drug additives. They were then starved for 1 day before any experiments with the antibiotics. For the next 3 days, the water in the three tanks was treated with an antibiotic – a different one per tank. The dosage of the antibiotic was adjusted to 100 mg kg^{-1} fish body weight per day. A fourth tank was used as a reference, and the fish in it were fed normal commercial food without any antibiotics. All the experimental conditions, including the temperature, volume and concentration of the drugs in each tank were kept constant throughout the experiment. After 3 days the drug dosing was discontinued, and the fish were sampled for analysis every fifth day

after cessation of the drug treatment, i.e. on days 5, 10, 15 and 20 after the treatment stopped.

2.3.2. Fish sample preparation

For each analysis, a weighed aliquot of fish muscle (3 g) was homogenised in 3.0 mL of 0.1 mol L^{-1} EDTA, and 2.0 mL of pH 4.0 McIlvaine buffer, with the use of a Model HY-4 homogenizer (Shanghai Instrumental Manufacturing, Shanghai), at 4000 rpm min^{-1} for 15 min. Such a sample was extracted with 3 mL of acetonitrile, and then shaken for 5 min. It was then, centrifuged at 4000 rpm for 10 min, decanted into a clean polypropylene tube, and the muscle residue was extracted with another 3.0 mL aliquot of acetonitrile. The two extracts were then combined and evaporated to dryness with nitrogen. Dried extract was reconstituted in methanol (200 μL), and was centrifuged under the same conditions as before and stored at 4 °C in the dark.

2.3.3. Animal feed preparation and ultrasonic extraction

Animal feed samples spiked with target analytes, were prepared by adding the composite stock solution of the three antibiotics with three spiked levels, 0.1, 0.2 and 0.4 mg g^{-1} for each drug, to the feed samples, and then the samples were shaken for 2 min by a Model HY-4 shaker (Shanghai) and mixed completely. They were subsequently evaporated to dryness in air and stored at 4 °C in the dark.

An animal feed sample (0.1 g) was weighed accurately and placed in a 100 mL shaking flask with a stopper. Then, a 5.0 mL aliquot of ethyl acetate was added and the sample was treated in an ultrasonic bath for 20 min. The supernatant ethyl acetate layer (3.0 mL) was transferred into a centrifuge tube with a stopper, and centrifuged for 10 min at 4000 rpm. The ethyl acetate extract was evaporated to dryness with flowing nitrogen; the dried material was dissolved with methanol (500 μL) and diluted to 10.0 mL with water in a volumetric flask. In contrast, for the extraction of the fish samples, only ethyl acetate was used, thus simplifying the procedure. Satisfactory results were obtained (Section 3.4).

2.4. Experimental procedure

A suitable amount of each antibiotic, together with 2.0 mL of pH 3.78 Britton–Robinson buffer were transferred to an electrochemical cell and diluted to 10.0 mL with distilled water. This solution was purged by purified nitrogen gas for 360 s; it was allowed to stand for 10 s, and then submitted to a pre-concentration step with stirring for 60 s at an HMDE with a deposition potential of −800 mV. After a 2 s static period, a differential pulse voltammetric scan was run from −800 to −1200 mV with a scan rate of

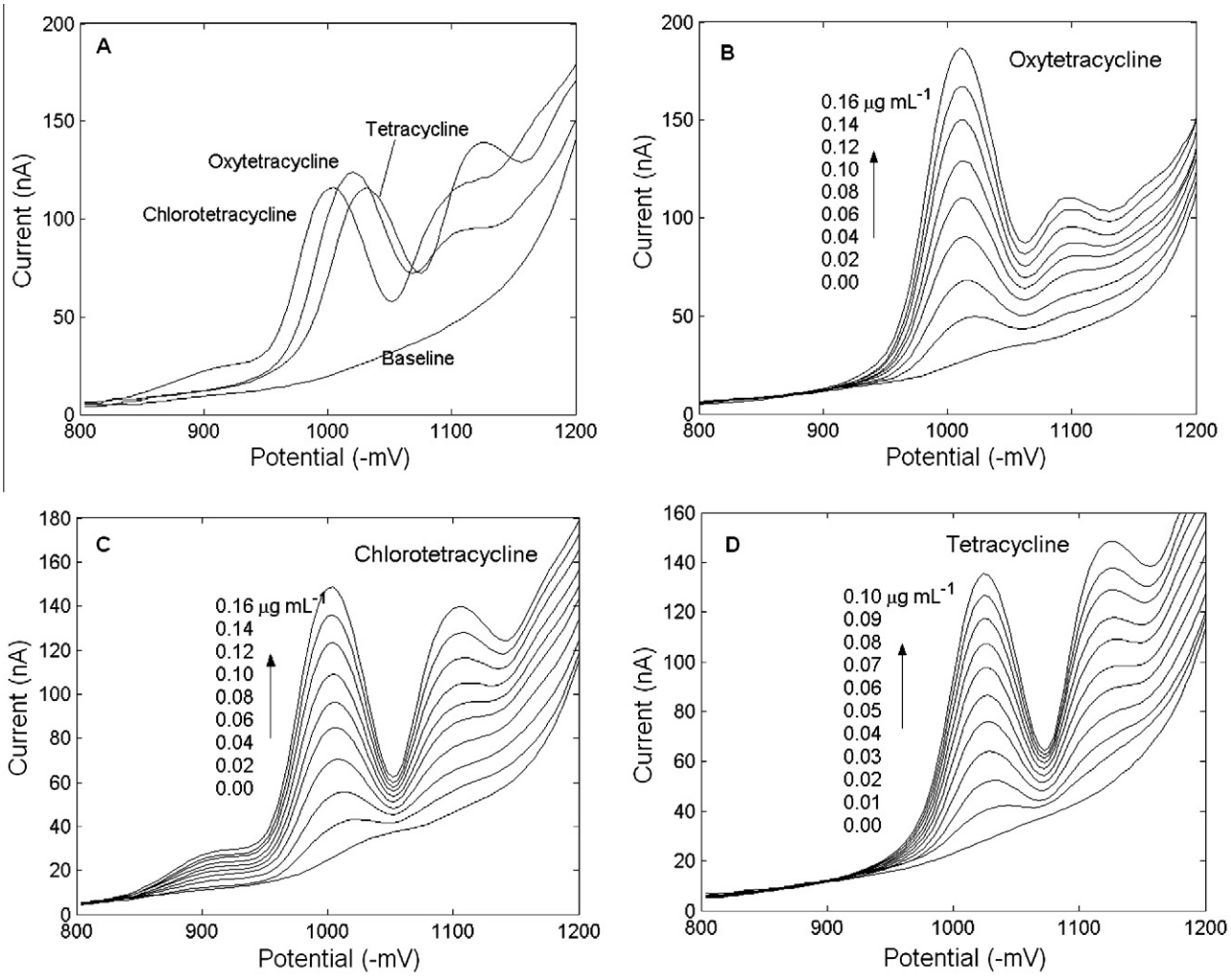


Fig. 1. Differential pulse stripping voltammograms: (A) chlorotetracycline (0.14 μg mL^{-1}), tetracycline (0.09 μg mL^{-1}) and oxytetracycline (0.11 μg mL^{-1}), and (B, C and D) three tetracyclines as a function of concentration (μg mL^{-1}) at a hanging drop mercury electrode immersed in Britton–Robinson buffer at pH 3.78. Scan rate = 20 mV s^{-1}, deposition potential = −800 mV and deposition time = 60s.

100 mV s^{-1}. The resulting voltammograms were measured and sampled in the range of −800 to −1200 mV at 4 mV intervals (total 101 potential points). The obtained voltammetric curves of each tetracycline (Fig. 1A) show that the voltammograms of these compounds heavily overlap.

2.5. Chromatographic procedure

The determination of OTC, TC and CTC drugs in animal feeds was verified by HPLC with detection at 375 nm (Huq, Garriques, & Kallury, 2006). Two solutions were used for the mobile phase: (A) acetonitrile and (B) 0.01 mol L^{-1} NaH_2PO_4 (adjusted to pH 3.0 with 10% H_3PO_4). A gradient was developed as follows: 15% (v/v) solution A was maintained between 0–5 min, then the solution A (v/v) ratio was brought upto 40% stepwise and then maintained at that level. The flow rate and temperature were kept at 1.0 mL min^{-1}, and 30 °C, respectively. Under these conditions, the retention times were 5.0, 5.8 and 9.8 min for OTC, TC and CTC analytes, respectively. Twenty microliters of each sample solution was injected, and the concentrations were calculated on the basis of peak area ratios.

2.6. Selection of a supporting electrolyte and the effect of pH

To find the optimum experimental conditions, several supporting electrolytes were tested including hydrochloric acid as well as a number of buffers: sodium citrate–hydrochloric acid, Britton–Robinson and boric acid–sodium hydroxide. It was found that the best defined peak waves and a relatively high sensitivity for the three antibiotics were observed with the use of the Britton–Robinson buffer.

The effect of pH (Britton–Robinson buffer) on the reduction peak current and peak potential of the voltammograms of the three analytes, OTC, TC and CTC, was investigated in the pH range 1.98–9.62. The results showed that all voltammetric peaks for the three TCs shifted to more negative potentials with increasing pH in a roughly linear manner. The peak current of the voltammograms of the analytes decreased with increasing pH. Taking into account the peak sensitivity and their resolution, a pH of 3.78 was selected as the supporting electrolyte.

2.7. Electrochemical experimental conditions

DPSV peak sensitivity increased with the increasing pulse amplitude. However, the quality of the peak shape deteriorated when the pulse amplitude exceeded 50 mV. Therefore, this value was chosen as the experimental pulse amplitude. In addition, the peak current decreased with the increasing pulse width. To obtain relatively high sensitivity and well-shaped waves, 50 mV and 40 ms were chosen for the pulse amplitude and width, respectively.

3. Results and discussion

3.1. Characteristics of the reduction process

According to the molecular structures of the TCs, the reduction process at the electrode should be affected by pH. In a moderately strong acidic medium, the dimethyl – amine group should be protonated (Wei & Lu, 1989), and the carbonyl group of ring A ortho-to the amine – N is easily converted to the hydroxyl group; two electrons and two protons are involved; with the application of the more negative potential, the carbonyl group of ring C is then reduced and a second reduction peak is observed. Thus, two common reduction peaks are observed for all three TCs and involve, in each case, a two electron change. From the cyclic voltammograms of the three TCs (Fig. 2), large and well-defined reduction peaks were observed, and no oxidation peaks were present. This indicated an irreversible electrode process for each compound. When the static period prior to the electrode reduction was increased, the intensity of the response also increased. This showed that the reduction peaks were related to the deposition/absorption processes of the analyte at the electrode.

3.2. Analysis of single tetracyclines: calibration models and limits of detection

The analyses were carried out as previously described (Sections 2.4 and 2.5), and linear calibration models were obtained (Table 1). It was found that the linear ranges were 0.02–0.16, 0.02–0.10 and 0.02–0.18 μg mL^{-1} for OTC, TC and CTC, respectively (Fig. 1B–D), and their corresponding LODs were 5.2, 4.1 and 3.5 μg L^{-1}. These results clearly indicated that the proposed electrochemical method was suitable for the determination of individual tetracyclines. However, no extensive clean-up procedure was used for the complex, spiked, real samples in this work and generally, a more complex background is expected. Thus, LOD values may be somewhat higher than otherwise.

These figures of merit were better than those obtained from a sensitive fluorescence method (Valverde et al., 2006a,b) (linear ranges were 0.08–4.0, 0.1–4.0 and 0.08–1.5 μg mL^{-1} for OTC, TC and CTC, respectively; corresponding LODs were 32, 14 and 5.0 μg L^{-1}, i.e. significantly worse to those above. Also, the method proposed in this work performed better in comparison with the HPLC one (linear range: 10.36–518.0, 12.11–605.5 and 11.79–589.5 μg mL^{-1} for OTC, TC and CTC, respectively (Wang, Peng, & Liu, 2008); corresponding LODs were 3.00, 3.97 and 8.46 μg mL^{-1} (a UV detector). Also, analysis with the resonance Rayleigh scattering (RRS) method (Wang et al., 2008) reported LODs of 2.12, 2.38 and 5.12 μg mL^{-1} measured with an RRS detector. Thus, generally, the proposed voltammetric method appears to perform better than the HPLC and fluorescence methods, and has a similar analytical capability to the more complex and costly RRS technique. However, as illustrated by the analysis of two quinolones in water samples (Reverte et al., 2003), the solid-phase extraction followed by the HPLS–MS determination, clearly provides better LOD (4–6 ng L^{-1}) and satisfactory Recoveries (88–112%). Hence, if available, this is the preferred technique for analysis of this type of compound including the antibiotics.

3.3. Chemometrics analysis of synthetic mixtures of tetracyclines

In voltammetric measurement of antibiotics, there is a proportional relationship between the peak current and the analyte concentration over a given concentration range:

$$i_j = k_{0j}c_0 + k_{1j}c_1 + k_{2jc_2} + \cdots + k_{ij}c_n = \sum_{i=1}^{n} k_{ij}c_i \quad (i = 1,2\ldots n;\ j = 1,2\ldots,s) \tag{1}$$

where, i_j, is the voltammetric current of the mixture at the jth potential point, c_i is the ith component in mixture, k_{ij} is the proportional coefficient for component, i, at potential point, j (total s points are selected), and $k_{0j}c_0$ ($c_0 = 1$) is the bias term responsible for the residual current, which varies with potential, and in general, cannot be neglected. By measuring the current at s potentials, s equations are obtained, which can be expressed in matrix form as:

$$\boldsymbol{i}_{1\times s} = \boldsymbol{c}_{1\times(n+1)}\boldsymbol{K}_{(n+1)\times s} \qquad (j = 1,2,\ldots,s) \tag{2}$$

where $\boldsymbol{i}$ and $\boldsymbol{c}$ are vectors. The coefficient matrix, $\boldsymbol{K}$, may be obtained by calibration of m standard samples and Eq. (2) can be expressed in matrix form:

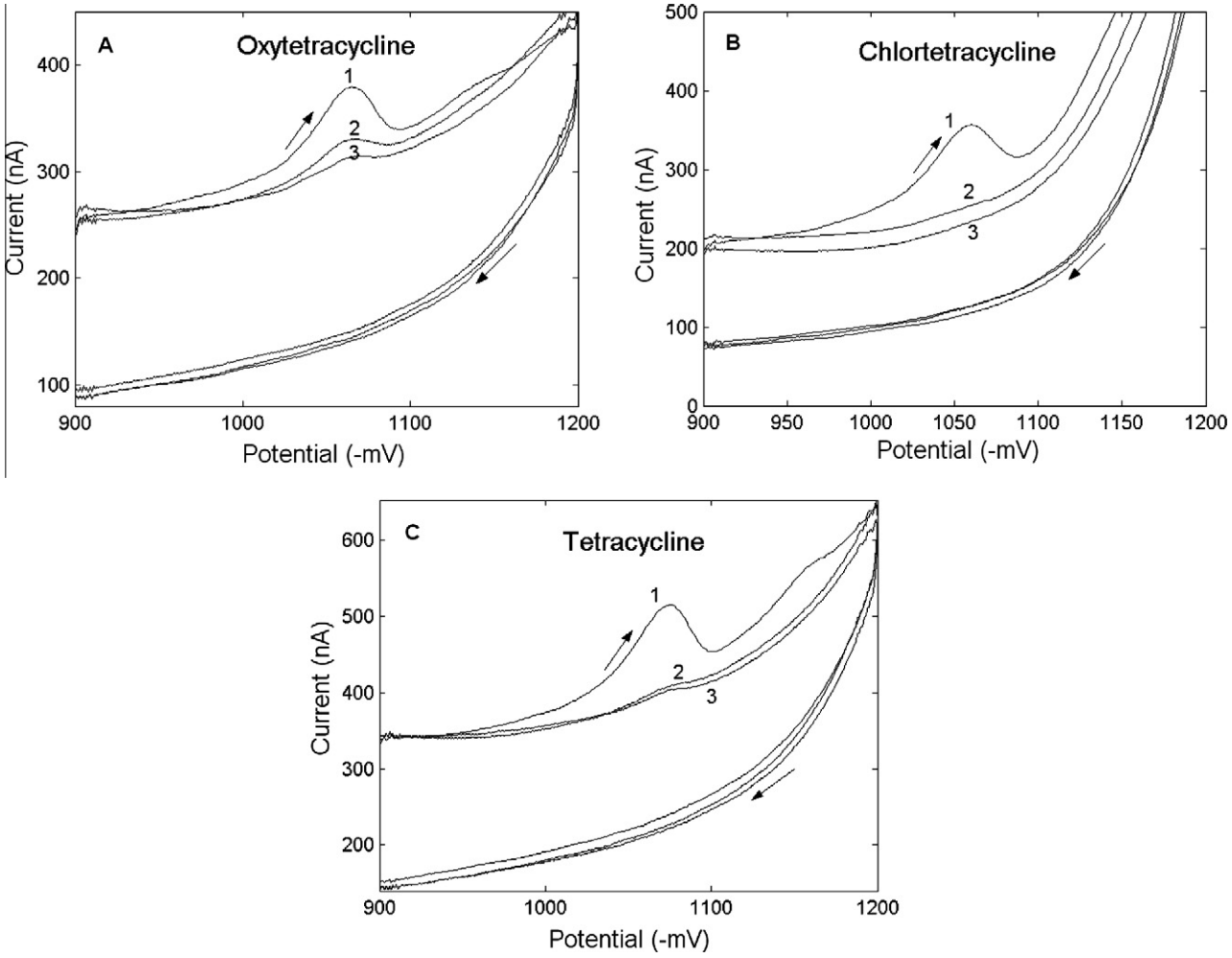


Fig. 2. Cyclic voltammograms of three tetracyclines solutions (each analyte – 0.15 μg mL^{-1}). Scan rate = 20 mV s^{-1}.

Table 1
Parameters of linear calibration models for oxytetracycline, chlorotetracycline and tetracycline.

Parameters	Oxytetracycline	Tetracycline	Chlorotetracycline
Sample number (n)	8	10	9
Linear range (μg mL^{-1})	0.02–0.16	0.02–0.10	0.02–0.18
Intercept of calibration curve (nA)	5.70	−1.22	3.93
Slope of calibration curve (μA mL μg^{-1})	0.881	0.777	0.570
Correlation coefficient	0.9992	0.9993	0.9993
S.D. of intercept	1.71	1.71	1.26
S.D. of slope	1.32	7.49	6.72
S.D. of regression	0.725	2.38	1.11
Limit of detection (μg L^{-1})	5.2	4.1	3.5

$$I_{m\times s} + C_{m\times(n+1)}K_{(n+1)\times s} \quad (3)$$

where the first row in the matrix, ***K***, represents the background vector. According to this equation, it is possible to determine the analytes individually by multivariate calibration methods. In this work, the electrochemical data were submitted to PLS and PCR modelling of the multi-component mixtures (Feudale et al., 2002; Geladi & Kowalski, 1986; Otto & Wegscheider, 1985).

The calibration data set for quantitative analysis of the three analytes in a mixture was designed with the aid of an orthogonal design method so as to have a statistically balanced model, which would reveal maximum information. A set of sixteen different samples was prepared according to a four-level orthogonal array, denoted by OA_{16} (4^5) (Lan, Wong, Chen, & Sin, 1994). For the mixed samples, the ranges of concentrations used for oxytetracycline, tetracycline, and chlorotetracycline were 0.03–0.07, 0.01–0.055 and 0.01–0.055 μg mL^{-1}, respectively. Each mixed sample was measured and recorded between −800 and −1200 mV by DPSV according to the experimental procedure, and the data matrix consisted of 16 rows (number of samples) and 104 columns – three corresponding to the concentrations of the analytes (independent variables, ***C*** matrix) and 101 columns of peak current data (dependent variables, ***I*** matrix). This matrix was pretreated by autoscaling, i.e. y-Mean-centering and variance-scaling. It was submitted to the PCR and PLS chemometrics methods. The optimal number of factors was selected for the modelling with the use of

leave – one – sample – out – at – a time method (Brereton, 2002). Three factors were found to be significant at the $p = 0.05$ level for both the PCR and PLS calibration models. The associated uncertainties were calculated for each calibration method with the use of the relative prediction error (RPE). For a single component in mixtures RPE was (Otto & Wegscheider, 1985):

$$\mathrm{RPE_S} = \left[\frac{\sum_{i=1}^{n}\left(c_{ij(\mathrm{found})} - c_{ij(\mathrm{added})}\right)^2}{\sum_{i=1}^{n}\left(c_{ij(\mathrm{added})}\right)^2}\right]^{0.5}$$

and the global (total) RPE for all components was:

$$\mathrm{RPE_T} = \left[\frac{\sum_{i=1}^{n}\sum_{j=1}^{m}\left(c_{ij(\mathrm{found})} - c_{ij(\mathrm{added})}\right)^2}{\sum_{i=1}^{n}\sum_{j=1}^{m}\left(c_{ij(\mathrm{added})}\right)^2}\right]^{0.5}$$

where $c_{ij,\mathrm{added}}$ indicates the concentration of jth component in ith mixtures and $c_{ij,pred}$ is its estimation found by the above mentioned chemometric methods.

Another sixteen ternary synthetic mixtures containing the three TCs were independently prepared for the verification of the calibrations. The PLS and PCR prediction models (three factors were selected for both PLS and PCR) performed well for verification: the $\mathrm{RPE_S}$ was in the range of 4.5–9.6% and the $\mathrm{RPE_T}$ was less than 6.7%, whilst % Recoveries ($=100 \times (c_{\mathrm{found}}/c_{\mathrm{added}})$) generally were in the range of 103–107% for mixtures of the three analytes. These results were comparable to the results obtained by simultaneous fluorescence determination of the TCs (% Recoveries: 86.7–111.0%, 92.0–115.0% and 75.0–116.7% for OTC, TC and CTC, respectively) (Valverde et al., 2006a,b), and also, similar to the HPLC method (% Recovery of 95.0–105.0% for synthetic and some commercial samples) (Wang et al., 2008).

3.4. Simultaneous determination of the three antibiotics in animal feeds

Commercially available animal feed was checked to be free of the antibiotics, and was spiked with the three drug analytes at 0.1, 0.2 and 0.4 μg g^{-1} levels. Such samples were then used for verification of the proposed analytical procedure (Section 2.4). The volammograms of the spiked feed sample (Fig. 3) shows that the peak current (curve 1) was somewhat different from the standard (curve 2), and was relatively low. The analyte extraction procedure referred to previously may have contributed to this observed difference.

The % Recovery results obtained from the DPSV data and the PLS calibration model (in the range of 81.3–87.9%), indicated that the analytical methodology was satisfactory. The % Mean Recoveries from the different samples by the proposed and the HPLC methodology were consistent. Thus, the proposed method performed satisfactorily for the simultaneous determination of oxytetracycline, tetracycline, and chlorotetracycline in animal feeds.

3.5. Analysis of fish samples

Samples of the carp tissue treated with the antibiotics as previously described (Sections 2.3.2 and 2.4), were analysed by the proposed method (Table 2). Fifteen days after withdrawal of the fish from the drug – spiked environment, the detected concentrations of drugs were lower than the MRL tolerances for TCs of the US and EU standards (Webs). The presented results are in general agreement with those from previous studies of other fish species (Malvisi, Della Rocca, Anfossi, & Giorgetti, 1996). There are many reasons, which can affect the drug residence time in fish, e.g., different water temperatures and the period of feeding. In this work, the high concentrations of TCs were particularly evident during the first few days after withdrawal of the fish from the drug environment, but, even after a fortnight, the drugs were still present at relatively high concentrations. If such fish were to be ingested regularly by humans for prolonged periods of time, there would be cause for concern to the health of the individuals involved.

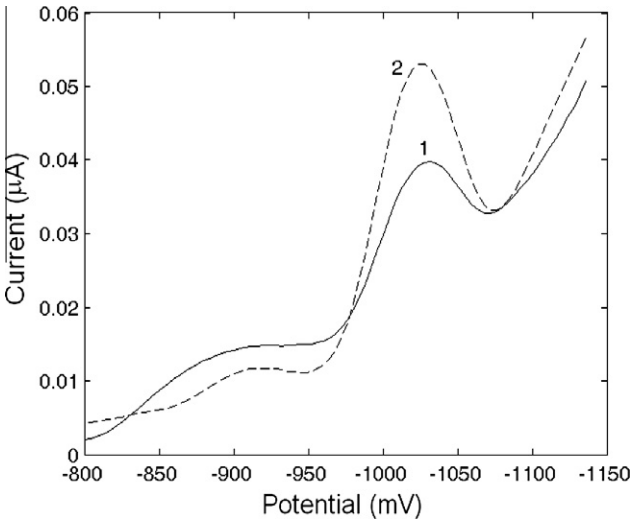


Fig. 3. Voltammograms of (1) the spiked feed samples (0.4 μg g^{-1} for each tetracycline); (2) a standard with same contents as 1.

Table 2
Tetracyclines residues in Carp fish tissue sampled after treatment with the antibiotics for three consecutive days followed by fish withdrawal from a fresh water tank after a specified time.

Compounds	Drug concentration after fish withdrawal at a specified time (μg kg^{-1}, $n = 5$)[a]			
	5 days	10 days	15 days	20 days
Oxytetracycline	248 ± 4[b]	112 ± 2	82 ± 4	35 ± 4
Chlorotetracycline	252 ± 5	135 ± 3	98 ± 5	45 ± 4
Tetracycline	220 ± 5	102 ± 4	76 ± 3	27 ± 6

[a] PLS was used with three factors.
[b] Mean ± standard deviation.

4. Conclusions

A simple, cost effective and well performing differential pulse stripping voltammetry method for simultaneous determination of three tetracycline antibiotics in different real food samples was researched and developed. It was verified with the aid of chemometrics PCR and PLS prediction modelling. The figures of merit were significantly better than those found for the alternative HPLC and fluorescence calibrations, and were also similar to those obtained with the resonance Rayleigh scattering technique. However, as anticipated, the method is less sensitive than the HPLC–MS one. Thus, in general, the proposed novel voltammetric procedure is highly competitive for the determination of these antibiotics in foods, particularly as a screening method.

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